

Homework #1

Introduction to Proofs

1. Factor $2^{15} - 1 = 32,767$ into a product of two smaller positive integers.

(a) Earlier in the book, there is a theorem that essentially states

$$xy = (2^b - 1) (1 + 2^b + 2^{2b} + \cdots + 2^{(a-1)b}),$$

where the number can be written as $2^n - 1$ and a and b are less than n .

I selected $a = 5$ and $b = 3$, and rewrote 32,767 as

$$(2^3 - 1) \left(\sum_{n=0}^{a-1} 2^{bn} \right).$$

This gives the two positive factors

$$(7)(4681).$$

(b) Find an integer x such that

$$1 < x < 2^{32,767} - 1$$

and $2^{32,767} - 1$ is divisible by x .

Using the same logic, we can find that one of the factors of this number is

$$2^7 - 1 = 127.$$

2. Make some conjectures about the values of n for which $3^n - 1$ is prime, or the values of n for which $3^n - 2^n$ is prime.

(a)

n	$3^n - 1$	Prime?	$3^n - 2^n$	Prime?
1	2	✓	1	×
2	8	×	5	✓
3	26	×	19	✓
4	80	×	65	×
5	242	×	211	✓
6	728	×	665	×
7	2186	×	2059	✓
8	6560	×	6305	×
9	19682	×	17707	✓
10	59048	×	58025	×

For all $n > 1$, $3^n - 1$ is not a prime

3. The proof of Theorem 3 gives a method for finding a prime number different from any in a given list of prime numbers
 - (a) $(2 * 3 * 5 * 7) + 1 = 211$
 - (b) $(2 * 3 * 5 * 7 * 11) + 1 = 2311$
4. Find five consecutive integers that are not prime.
24,25,26,27,28
5. Use the table in Figure I.1 and the discussion on p. 5 to find two more perfect numbers
 - (a) 31 and 127
6. The sequence 3, 5, 7 is a list of three prime numbers such that each pair of adjacent numbers in the list differ by two. Are there any more such “triplet primes”?
 - (a) If a number is divisible by 3 then $\text{mod}(n, 3) = 0$, however, mod is a cyclic function and will either be 0,1,2. with leave only 2 other number that can be factors, breaking the sequence of 3. the only sequence that satisfies this is the one given.
7. A pair of distinct positive integers (m, n) is called amicable if the sum of all positive integers smaller than n that divide n is m, and the sum of all positive integers smaller than m that divide m is n. Show that (220, 284) is amicable.
 - (a) 220's factors that are less than 220 are 1,2,4,5,10,11,20,22,44,55,110. The factors add up to 284, so it is amicable.